JahMari Harrigan

Test and Evaluation

Automated integration tests ran against actual MySQL 8.4.7 and an HTTP Node.js server. These results evaluate technical behavior; they are not participant feedback or a substitute for browser usability testing. The same sixteen tests also passed after creating a fresh database. Detailed executable steps and assertions are in tests/system.test.js.

Test	Requirement or scenario	Actual result	Evidence
T01	Registration and login	PASS	production-tests.txt
T02	Valid empty and invalid search	PASS	production-tests.txt
T03	Authentication and CSRF rejection	PASS	production-tests.txt
T04	Server fare and complete persistence	PASS	production-tests.txt
T05	Customer ownership	PASS	production-tests.txt
T06	Concurrent same-seat requests	PASS	production-tests.txt
T07	Multi-passenger rollback	PASS	production-tests.txt
T08	Cancellation and inventory release	PASS	production-tests.txt
T09	Malformed passenger and seat	PASS	production-tests.txt
T10	Administrator authorization	PASS	production-tests.txt
T11	Immutable booked fare	PASS	production-tests.txt
T12	Analytics reconciliation	PASS	production-tests.txt
T13	Airport and aircraft maintenance	PASS	production-tests.txt
T14	Full capacity and schedule cancellation	PASS	production-tests.txt
T15	Logout revocation	PASS	production-tests.txt
T16	MySQL FK and check constraints	PASS	production-tests.txt

T17 separately passed server-restart persistence; its saved output is evidence/persistence-check.txt. Tests use fictional accounts and remove their own test bookings afterward.

Critical Integrity Test Procedures

T06 concurrency	Two authenticated users submit the same free seat concurrently. Expect one HTTP 201, one HTTP 409 and one active allocation.	Passed: one accepted request and one conflict; count equals one.
T07 rollback	Submit two passengers: first requests a free seat; second requests an occupied seat. Expect failure with unchanged booking and passenger counts.	Passed: no partial booking or passenger records.

T08 cancellation	Cancel an owned confirmed booking. Retrieve status, inspect payment, and query seat availability.	Passed: cancelled, voided and seat available.
T11 fare change	Save booking totals, edit departure fare, read totals again, restore fare.	Passed: existing totals unchanged.
T14 full flight	Create a future departure; fill all 24 seats in groups of four; try another allocation; cancel departure.	Passed: extra booking rejected; all active seats released and bookings cancelled.
T17 restart	Create booking and session, stop Node.js, start it again, retrieve and cancel with saved session.	PASS: booking, passenger, price and session survived restart; cancellation then succeeded.

Defects and retests

A production test run failed because a database process in a different sandbox held the data directory and was unreachable. The preview context was stopped and the same suite was rerun in a shared execution context; all sixteen tests passed. This was not counted as a successful run. The initial test for rejected database writes was reviewed to ensure database connectivity had already been proven by the preceding tests.

Owner walkthrough and remaining checks

The owner completed a guided browser walkthrough on October 10, 2026. Screenshots support booking, administrator access, fare edits, analytics and printing. Seat release, login retrieval, narrow-window layout and keyboard navigation were reported by the owner. Evidence follows. Actual mobile-device and assistive-technology checks remain unverified.

Three User Evaluation Records

Three participants completed the tasks listed below using fictional flight and payment data. Participant 3 also reviewed administrator features. The final testing session took place on October 25, 2026.

Task	Record
Search a flight	Route/date chosen; completion; confusion; elapsed time if measured
Create a reservation	Passenger entry; seat selection; errors; assistance needed
Retrieve confirmation	Reference retrieval; clarity of fare/status; completion
Cancel reservation	Completion; understanding of released seats; error messages
Administrator review by at least one user	Create or edit a schedule; inspect reservation; role clarity

Ratings and feedback

Ratings use a scale from 1 (poor) to 5 (excellent) for ease of use, clarity, reliability and overall satisfaction. The participant table records the supplied ratings, browser, device and feedback for each session.

JahMari Harrigan

Participant	Session status	Actual ratings and comments
P01	Owner-reported session: October 9, 2026	Ratings, feedback and session details are recorded in the participant feedback section.
P02	Owner-reported session: October 11, 2026	Ratings, feedback and session details are recorded in the participant feedback section.
P03	Completed October 25, 2026	Administrator checks completed; ratings and feedback are recorded in the participant feedback section.

The owner-supplied session records list Participant 1 on October 9, 2026, Participant 2 on October 11, 2026, and Participant 3 on October 25, 2026. Participant 2 retested the navigation improvements and completed the other tasks. Detailed results appear in the participant feedback section.

Evaluation Against Success Criteria

Criterion	Current evidence	Status
Correct fare calculation	T04 and T11	Verified by automated tests
Persistent bookings	MySQL integration plus restart check output	Verified by saved T17 PASS output
Duplicate-seat prevention	T06 plus database unique constraint	Verified
Atomic multi-passenger booking	T07 unchanged row counts after conflict	Verified
Cancellation consistency	T08 and T14	Verified
Role and ownership separation	T03, T05, T10 and T15	Verified
Fresh environment setup	clean-setup.txt	Verified on new database
Guided customer/admin browser flows	Owner walkthrough and screenshots on following pages	Observed/reported; limited to listed scenarios
Three real users complete tasks	Three owner-supplied session records, ratings and feedback	All three sessions reported complete; final session October 25

How to close the evaluation

For each real issue, add a defect ID, reproduction steps, expected and actual behavior, severity, correction and retest evidence. Update the implementation change log and affected requirements when a fix changes behavior. Do not replace earlier evidence silently. If the release changes, create a new commit and record the tested commit hash. Summarize remaining limitations and compare participant task completion with the original success criteria before submission.

JahMari Harrigan

Owner Walkthrough Results

JahMari Harrigan performed this guided owner walkthrough on October 10, 2026 using the local application. Browser version and device model were not collected. No elapsed times or numerical ratings were collected. This is separate from the three required user evaluation sessions.

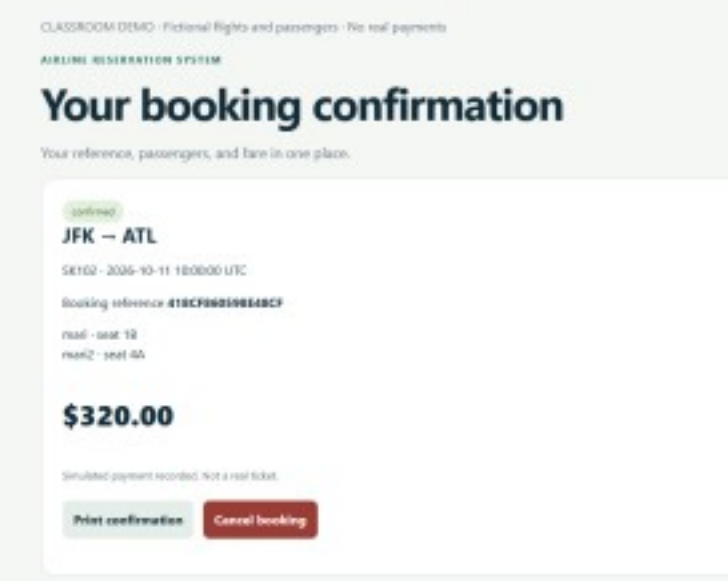
Check and expected behavior	Observed result	Evidence
M01 Book two seats at $160 each	Confirmed two passengers; saved total $320. PASS.	owner-booking.png
M02 Booking reduces seats; cancel releases them	Owner reported 22/24 after booking, 24/24 after cancellation. PASS by report.	Conversation; no cancellation screenshot
M03 Retrieve after logout/login	Owner reported booking remained retrievable. PASS by report.	Conversation; restart covered separately by T17
M04 Customer denied admin data; admin allowed	Denial screenshot and subsequent operations overview. PASS for observed pages.	owner-admin-denied.png; conversation
M05 Change SK102 fare to $175	Search showed the new $175 fare. PASS.	owner-fare-edit.png
M06 Analytics after one cancelled and one confirmed booking	2 reservations; 50% cancelled; $350; JFK ATL 0.3%; zero integrity mismatches. PASS.	owner-analytics.png
M07 Print retains details and hides search	CLI fix plus final preview showed details on one page with search hidden. PASS.	cli-print-after.png
M08 Narrow window and keyboard Tab/Enter	Owner reported both worked. PASS by report; not a full accessibility audit.	Conversation

No new automated suite was run during this walkthrough or report update. Existing automated logs remain unchanged. A 24-seat result initially viewed was for the reverse route, not evidence of an inventory defect.

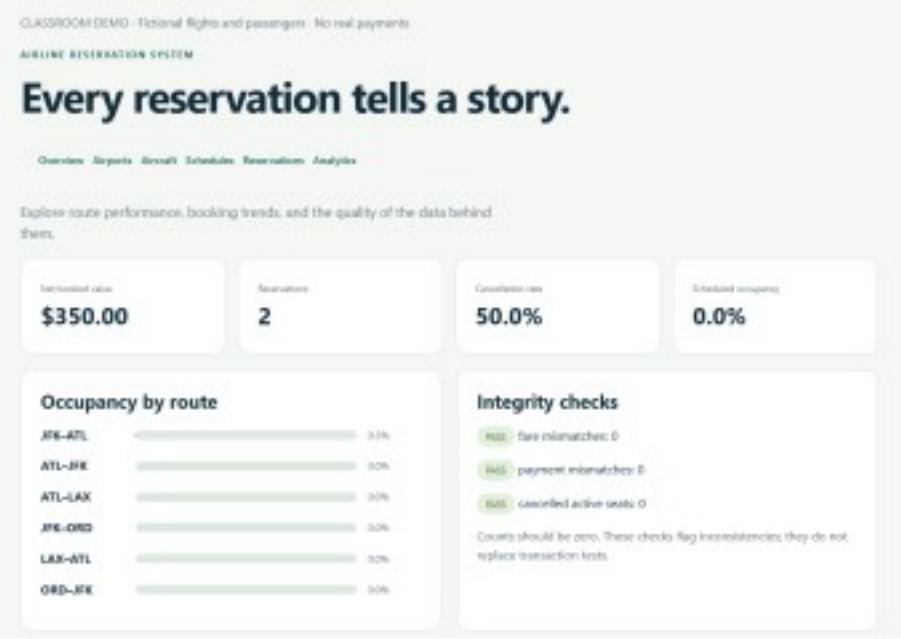
Booking and Analytics Evidence

Initial two-passenger confirmation at $160 per passenger. The later fare edit applied to new bookings.

JahMari Harrigan



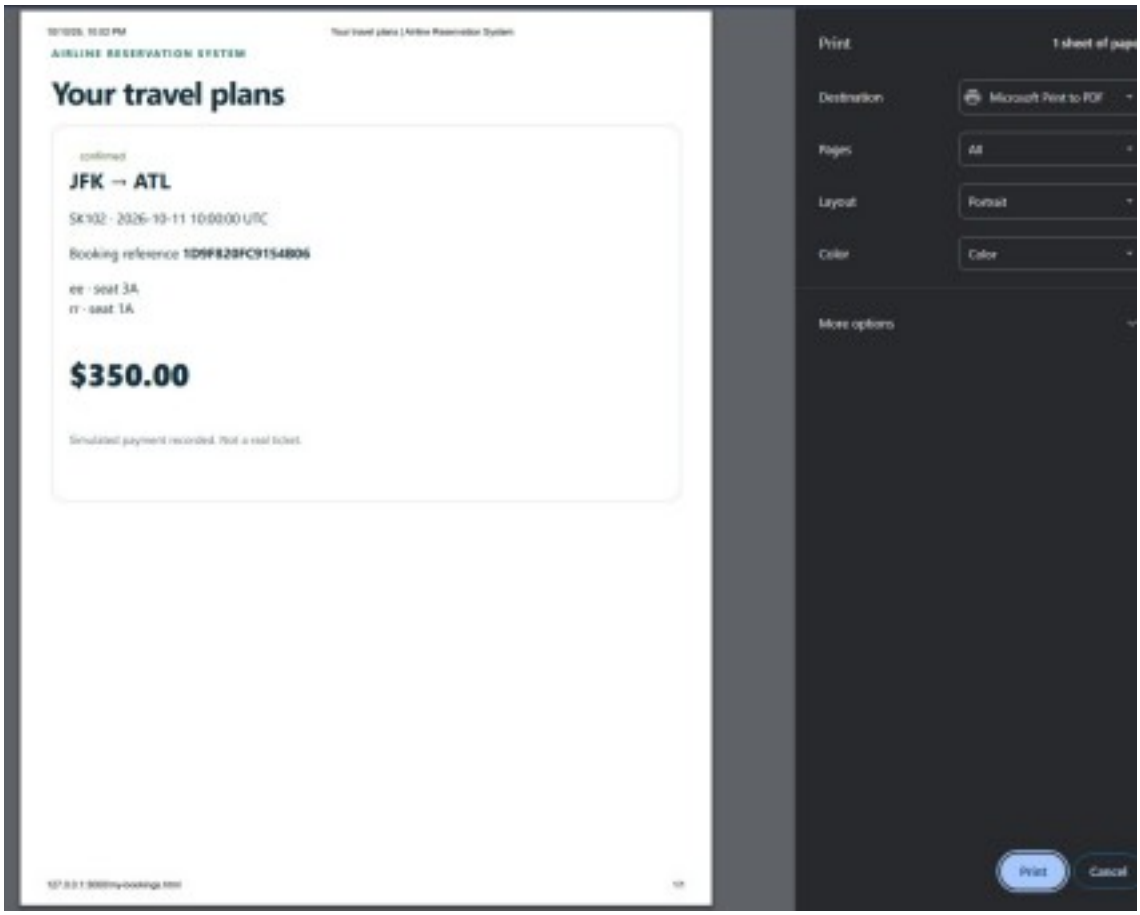
After cancellation and a new two-passenger booking at $175, the dashboard reconciled to $350 and 50% cancellations. Overall 0.0% is rounding of 2 occupied seats out of 4,320 scheduled seats; JFK-ATL is 2/720, displayed as 0.3%.



Print Defect and CLI Retest

PRINT-01: My Bookings print preview retained the lookup form and Show all link. The CLI added #lookup to the existing print only hiding rule. Five source checks passed. The owner refreshed the page and supplied the preview below; lookup controls are absent and all booking details remain visible. No physical print or saved PDF was tested.

JahMari Harrigan



Prompt, changes, environment errors and static results: evidence/cli-print-review.md. Browser verification was performed by the owner and reviewed in desktop Codex after the CLI session. This does not retroactively change the earlier desktop development into CLI work.

JahMari Harrigan

Participant Feedback and Reported Retests

Participant 1 reported vague one-word navigation tabs on October 9. The wording was clarified that day, and Participant 2 confirmed the improvement and completed the other tasks on October 11. Participant 3 completed the final testing session on October 25. The administrator test covered changing SK101 to $255, checking the updated search price and verifying that a customer account could not access the admin panel. Session results and ratings were supplied by the owner.

Participant and reported session date	Browser and device	Rating supplied	Feedback supplied by owner
Loni 10/09/2026	Google Chrome Laptop	Ease Of Use: 3 Clarity: 2 Reliability: 3 Overall Satisfaction: 3	One-word navigation tabs did not clearly explain their destination pages. Tasks worked. The wording was clarified on October 9 and retested by Participant 2 on October 11.
Jordan 10/11/2026	Google Chrome Laptop	Ease Of Use: 5 Clarity: 5 Reliability: 5 Overall Satisfaction: 5	Successful run. Confirmed the clearer navigation labels and completed the other tasks on October 11. Rated all four categories 5/5.
Isaac 10/25/2026	Opera Laptop	Ease Of Use: 5 Clarity: 5 Reliability: 5 Overall Satisfaction: 5	Successful admin test: changed SK101 to $255, checked the updated search price and confirmed customer access was denied. All four ratings were 5/5.

Issues and follow-up

USER-01: navigation and page context. The owner supplied “Flight” and “Find a flight” as examples when describing earlier wording changes, along with the notice “CLASSROOM DEMO · Fictional flights and passengers · No real payments” and “Built to be understood.” followed by “A systems analysis and data analytics project using fictional flights, transparent fares, and simulated payments.” These are recorded as owner-reported improvements, not requests for new interface changes. The owner reports Participant 2 confirmed improved clarity and attributes Participant 2 and Participant 3 ratings of 5/5 to it.

Current source verification: navigation reads "Find a flight"; the classroom demo notice and descriptive homepage text are present. The owner reports that the wording improvements followed the October 9 feedback and were confirmed by Participant 2 on October 11. A mistaken documentation-follow-up change to "Flight" was reverted. No booking-saving defect is attributed to Participant 1.